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Patient-independent EEG seizure detection using channel attention and dual-branch time–frequency features

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Abstract
We developed a patient-independent EEG-based seizure detection model that generalizes across subjects despite inter-subject variability in spatial seizure expression, electrode characteristics, and artifacts. Scalp EEG recordings from the CHB–MIT (18 channels) and the TUSZ v2.0.3 (20 channels) datasets were evaluated under strict subject-disjoint protocols, employing leave-one-subject-out (LOSO) cross-validation for CHB–MIT and the official subject-disjoint *train/dev/eval* split for TUSZ. A channel-wise attention front end adaptively reweights EEG channels prior to feature extraction. The reweighted signals are processed through dual-branch time–frequency representations based on empirical mode decomposition (EMD) and discrete wavelet transform (DWT). Lightweight attention-augmented convolutional neural networks model each branch, and their embeddings are fused for seizure/non-seizure classification. Episode-level decisions are obtained by averaging window-level probabilities from short, overlapping segments. The proposed model demonstrated robust cross-subject generalization under strict subject-disjoint evaluation on both datasets. Ablation studies showed that the EMD and DWT branches provide complementary information and that the complete configuration, combining front-end channel attention with intra-CNN attention, achieved the highest accuracy, AUROC, F1-score, and recall on both datasets. These findings indicate that complementary EMD–DWT representations combined with channel reweighting enhance generalization for patient-independent seizure detection. The proposed framework may support scalable and robust seizure detection across unseen patients and heterogeneous EEG acquisition settings.

Keywords Electroencephalogram · Patient-independent seizure detection · Channel attention · Healthcare access

1 Introduction

Epilepsy is a long-term brain disorder defined by repeated seizures that occur without a clear trigger, affecting tens of millions of people around the world. Seizures arise from bursts of abnormally synchronized neuronal activity and can appear very differently across patients, both on EEG and clinically, ranging from brief sensory

changes to motor events and altered awareness. Careful documentation of when seizures occur and how often they happen is vital for diagnosis, presurgical planning, and adjusting therapies. Yet in practice, reviewing many hours of scalp EEG by hand is slow, labor intensive, and varies between readers, especially in long inpatient and ambulatory recordings [1–3].

Automated seizure detection has become increasingly important as continuous EEG recordings can span hours or days, generating much more data than clinicians can review in real time. Manual annotation is slow, subjective, and often inconsistent across readers, which can delay diagnosis and limit quantitative analysis. Automated systems help to identify possible seizure events, guide clinicians to the most relevant parts of long recordings, and provide consistent, objective metrics in both inpatient and ambulatory settings. In practice, automated methods serve as a complement to clinical review, improving efficiency, scalability, and reliability [2, 3].

The main challenge is variability. In addition to variations in montage, hardware, sampling rate, and electrode impedance, ictal patterns can vary with age and syndrome even when recordings are acquired using similar hardware, and labels in large clinical datasets can be noisy or incomplete. Models trained on specific patients rarely transfer; practical systems must work across subjects and hospitals while keeping false alarms low and misses rare. These realities call for subject-disjoint evaluation and architectures that adapt spatial emphasis and remain robust to artifacts and inter-patient differences [2–4].

Machine learning and deep learning make scalable, consistent seizure detection possible by extracting reliable cues from non-stationary EEG or by learning spatio-temporal patterns directly [5, 6]. Recent CNN models with attention (e.g., squeeze–excitation, CBAM, and self-attention) report good accuracy in many settings [7–9]. However, many studies still rely on patient-specific or mixed-subject splits, and performance often drops when models are tested under subject-disjoint protocols on CHB–MIT and TUSZ [10, 11], highlighting unresolved cross-subject variability [2, 12]. Guided by these needs, this study aims to improve cross-subject generalization under strict subject-disjoint evaluation.

In this work, we learn a channel-wise attention gate for each 1 s window to adapt spatial emphasis across channels. From the reweighted signal, we compute two complementary time–frequency summaries—EMD-based IMF statistics and DWT sub-band statistics—and model each with a lightweight CNN that includes feature-wise and spatial attention [6, 9]. The two branch embeddings are fused for classification, and *episode-level* decisions are obtained by averaging per-window probabilities (soft voting) [13, 14].

Under subject-disjoint evaluation on CHB–MIT using leave-one-subject-out (LOSO) cross-validation and TUSZ v2.0.3 using the official split, the method achieves competitive patient-independent performance. Ablations confirm the contributions of the front-end channel attention, dual-branch EMD–DWT fusion, and intra-CNN attention [5, 12, 15–20]. A high-level summary of the full pipeline is shown in Fig. 1, with detailed component illustrations provided in Figs. 2, 3 and 4.

Recent hybrid seizure-detection approaches have increasingly combined complementary representations and learning architectures to improve EEG modeling. For example, multiple CNN branches have been used to jointly model time-domain, frequency-domain, and time–frequency representations [21], while convolutional and recurrent architectures have been integrated with attention mechanisms to capture both local and temporal information [22]. CNN–Transformer architectures have also been investigated for patient-independent seizure detection [23]. More recently, multi-branch graph-based models have combined different EEG connectivity representations through multi-head attention [24], and patient-adversarial learning with spatio-temporal augmentation has been explored to improve cross-patient generalization [20].

Despite these advances, several challenges remain for patient-independent seizure detection, particularly in adapting to subject-dependent variations in the spatial distribution of ictal activity while preserving complementary time–frequency characteristics of the EEG signal. To address these aspects, the proposed framework applies

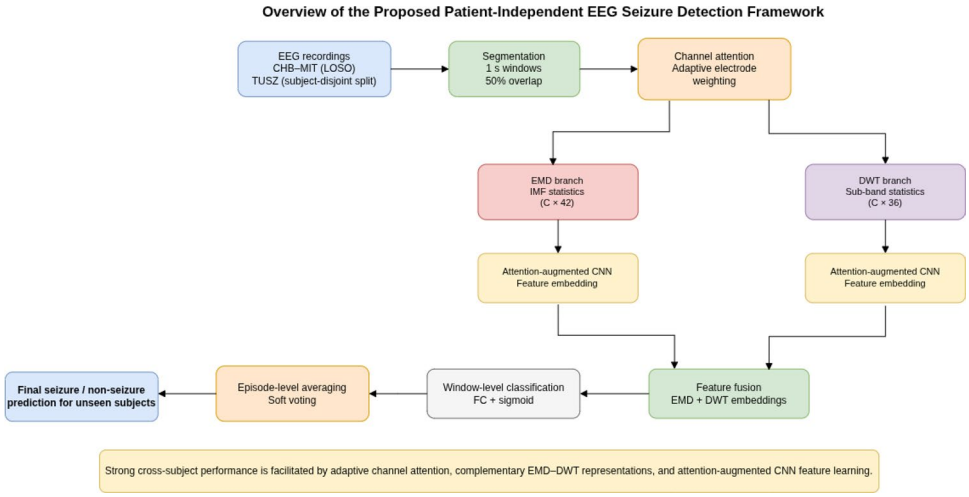


Fig. 1 Overview of the proposed patient-independent seizure detector. A 1 s EEG window is first reweighted using a channel-attention front end. The weighted signal is processed through two complementary time–frequency branches based on EMD-derived IMF statistics and DWT sub-band statistics. Lightweight attention-augmented CNNs model both branches, and the resulting embeddings are fused for window-level seizure/non-seizure classification. During inference, window probabilities within each episode are averaged to obtain the final episode-level decision.

channel-wise attention directly to the input EEG before time–frequency feature extraction, allowing the contribution of individual EEG channels to be adapted for each EEG window. The reweighted signal is subsequently represented using two different decomposition principles: data-adaptive EMD and fixed-basis multi-resolution DWT. These representations are modeled independently through separate attention-augmented CNN branches and combined through late fusion, thereby preserving decomposition-specific information before classification. Because this reweighting is applied once at the input, the same channel-weighted EEG representation is supplied to both decompositions; the input weighting is not learned independently within each branch. In addition, the complete framework is evaluated under strict subject-disjoint protocols on both CHB–MIT and TUSZ, together with episode-level aggregation and continuous-recording evaluation. Thus, the proposed design is distinguished by the combination of input-level adaptive channel reweighting, complementary EMD–DWT representations with branch-specific modeling and late fusion, and patient-independent evaluation across two heterogeneous public EEG datasets.

Overall, the work has the following contributions:

1. Channel-attentive front end that adaptively reweights EEG channels, mitigating inter-subject variability before feature extraction;
2. Dual-branch EMD–DWT feature maps with attention-augmented CNN modeling and late fusion, yielding a compact and expressive representation;
3. Patient-independent evaluation on CHB–MIT and TUSZ with segment-to-*episode* aggregation, demonstrating competitive cross-subject performance and validating each design choice via ablations.

The remainder of this paper is organized as follows. Section 2 reviews related work, from classical time–frequency methods through deep architectures, attention mechanisms, and patient-independent evaluation protocols. Section 3 (Methodology) details the datasets, preprocessing pipeline, and the model architecture. Section 4 explains training and inference protocols. Section 5 describes results and ablations. Section 6 discusses the findings, their clinical implications, and the limitations of the study. Section 7 concludes the paper.

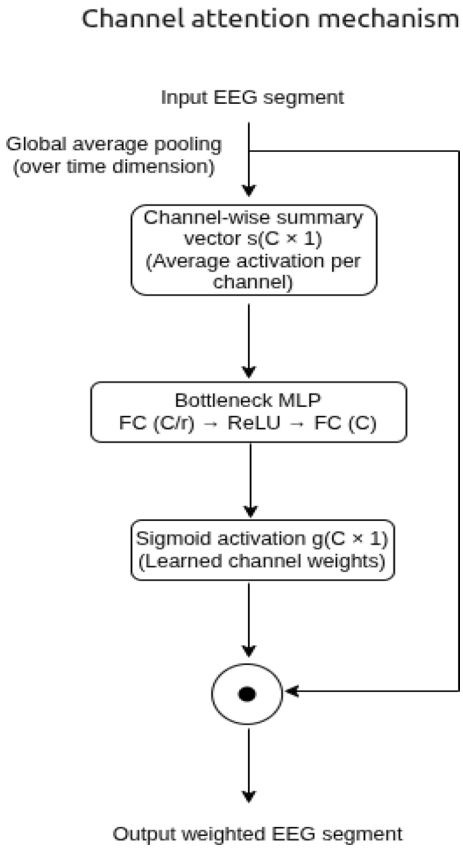


Fig. 2 Channel attention front end. For each input window $X \in \mathbb{R}^{C \times N}$, global average pooling over time produces a channel summary s. A bottleneck MLP ($C \to C/r \to C$) with ReLU and sigmoid yields channel gains g, which are used to reweight X and produce $\tilde{X}$, adapting spatial emphasis before feature extraction.

2 Related work

2.1 Classical time–frequency methods

Early automated seizure detectors typically followed a hand-crafted pipeline: transform each EEG channel into a time–frequency representation, summarize it with a compact set of statistics, and classify the concatenated features with a conventional learner. Among transforms, the discrete wavelet transform (DWT) is especially influential, providing a multiresolution analysis that is well matched to non-stationary EEG; it captures brief high-frequency transients and longer low-frequency rhythms via a dyadic filter bank [15–17, 25]. Typical pipelines compute per-band energies, entropies, and higher-order moments for each channel, then classify with SVMs, ANNs, or ELMs [15–17]. As a data-adaptive alternative, the empirical mode decomposition–Hilbert–Huang transform (EMD–HHT) decomposes the signal into intrinsic mode functions (IMFs), and the Hilbert transform produces instantaneous amplitude and frequency for each IMF, producing an adaptive time–frequency representation that complements fixed-basis wavelets [18]. Although effective and interpretable, these classical approaches generally treat all EEG channels uniformly, which can limit robustness when the channels carrying ictal evidence vary across patients [2–4].

From an application perspective, EMD and DWT remain complementary for seizure detection. EMD provides adaptive, signal-dependent oscillatory components, whereas DWT provides structured multi-resolution frequency information. Their joint use is therefore useful for representing seizure-related EEG patterns that may appear as both irregular local oscillations and multiscale frequency changes.

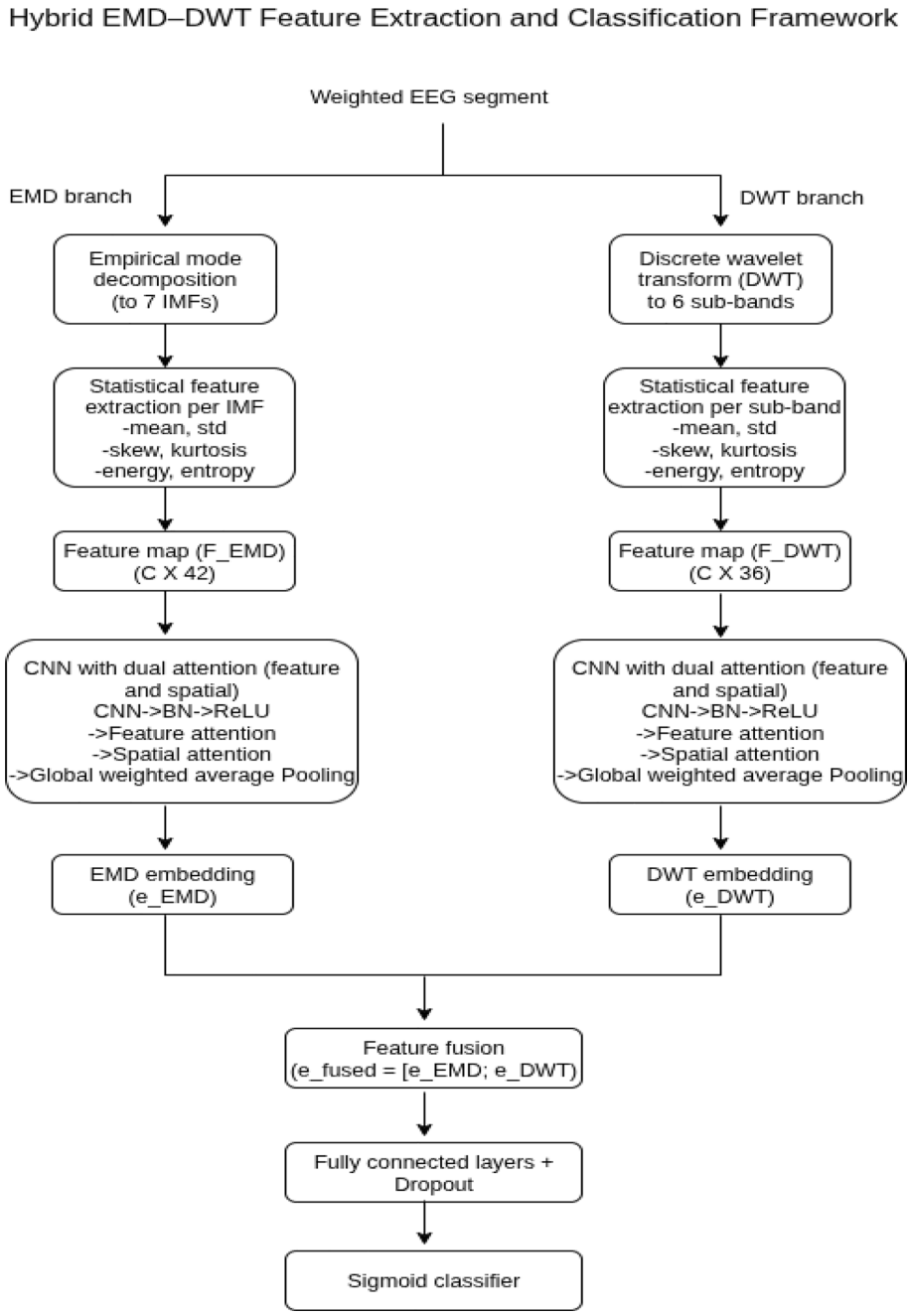


Fig. 3 Dual-branch EMD–DWT feature extraction and fusion. The weighted EEG segment is decomposed into two parallel feature sets using EMD and DWT. Each branch computes statistical descriptors to form feature maps of size $C \times 42$ (EMD) and $C \times 36$ (DWT), which are modeled by attention-augmented CNNs with global weighted average pooling. The resulting embeddings are concatenated and classified using a fully connected sigmoid layer.

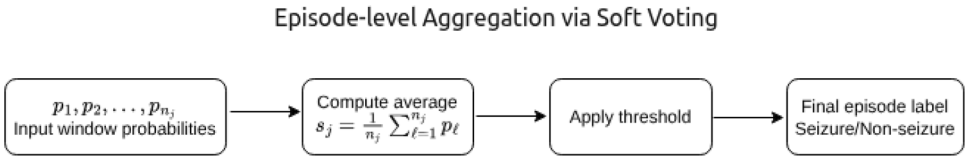


Fig. 4 Episode-level aggregation at inference. Window-level probabilities within each contiguous episode are averaged to obtain an episode score s_j, which is thresholded to assign the final label. This stabilizes decisions and reduces isolated window errors.

2.2 Deep learning architectures

Deep neural networks shift from hand-crafted features to learned spatio-temporal filters. Compact architectures such as EEGNet capture EEG rhythms efficiently using temporal and depthwise-separable convolutions [5], and deeper CNNs or hybrids have shown promise for end-to-end seizure detection and diagnosis [6]. Building on these backbones, hybrid designs that combine convolutional, recurrent, and Transformer components have been widely explored for seizure detection, including under patient-independent protocols [21–23].

Recent deep learning architectures such as Vision Transformers (ViT) [26], Swin Transformers [27], and DenseNet [28] have shown strong representation-learning ability in large-scale image and signal-analysis tasks. These models are important modern benchmark architectures; however, they were originally developed as general-purpose image backbones and are typically much larger than the compact EEG-specific model considered in this study.

2.3 Attention and EEG-channel-level modeling

Attention mechanisms improve representation by emphasizing salient patterns; feature-wise recalibration (e.g., squeeze–excitation) adapts the importance of intermediate feature maps, and spatial attention highlights informative regions; combined modules such as CBAM deploy both [7, 8]. Recent studies explore self-attention or hybrid attention blocks to improve discrimination [9]. Attention-based mechanisms have also been employed in other EEG deep-learning applications, including EEG-conditioned visual reconstruction, where attention is used to enhance learned representations [29]. Most attention mechanisms operate on learned feature maps within the network. In contrast, channel-wise gating at the input is less explored, particularly under patient-independent evaluation.

2.4 Transformer and foundation models

More recently, Transformer-based architectures, self-supervised learning, and foundation-model approaches have gained increasing attention in EEG analysis. Models such as BIOT [30] employ Transformer-based representations for cross-dataset EEG learning, while large-scale foundation models such as LaBraM [31] aim to learn generic EEG representations through self-supervised pretraining on diverse datasets. In addition, self-supervised representation learning approaches such as BENDR [32] and BrainBERT [33] demonstrate the usefulness of contrastive and masked modeling strategies for learning robust and generalizable EEG embeddings.

2.5 Hybrid and ensemble approaches

Recent studies across different application domains demonstrate the value of hybrid and ensemble learning for combining complementary modeling capabilities. For example, hybrid CNN–LSTM architectures combine convolutional feature extraction with sequential modeling, while hybrid VGG19–U-Net frameworks integrate deep semantic representations with spatial reconstruction capabilities [34, 35]. Attention-enhanced hybrid models such as SE–ResNet50 further demonstrate the use of channel-wise feature recalibration within deep networks [36]. In parallel, stacking and reliability-weighted fuzzy ensemble strategies have been used to combine heterogeneous model outputs and exploit complementary predictor strengths [37, 38]. Although these studies address different data modalities and application domains, they illustrate the broader methodological value of combining complementary representations, attention mechanisms, and model outputs within unified learning frameworks.

Alongside large deep-learning backbones, another relevant direction is to combine structured mathematical feature descriptors with neural networks for medical signal and image analysis.

For example, moment-based representations such as quaternion orthogonal Rademacher–Fourier moments, Meixner moments, and Charlier moments have been integrated with convolutional neural networks for tasks

including medical diagnosis, signal reconstruction, and visual speech recognition [39–42]. These approaches demonstrate that combining structured feature representations with deep learning can improve interpretability and robustness. Although these methods are not specifically designed for EEG seizure detection, they provide valuable methodological insights that motivate hybrid feature extraction strategies in our framework.

2.6 Evaluation protocols and research Gap

Across the seizure-detection literature, reported gains are frequently obtained under patient-specific or mixed-subject splits, which tend to overestimate generalization [2, 3]. Evaluation protocols therefore play a critical role. Surveys note that random splits can leak subject-specific patterns into both train and test sets, whereas leave-one-subject-out (LOSO) or other subject-disjoint protocols provide a more realistic measure of robustness [2–4]. Most prior work is evaluated on the CHB–MIT database and the TUH Seizure Corpus (TUSZ) [10, 11, 43]. Under subject-disjoint evaluation on these datasets, methods that appear strong under mixed-subject splits often show substantially lower performance, underscoring the challenge of patient-independent detection [2, 12].

Several recent directions aim to improve cross-subject robustness. Self-attention architectures, spiking neural networks, domain-shift mitigation, and cross-site training have been explored to reduce the performance drop observed under subject-disjoint evaluation [9, 12, 19, 20]. These studies indicate that patient-independent seizure detection requires EEG representations that remain stable across subjects, recording sites, and acquisition conditions. Compact hybrid representations also remain attractive because they can preserve domain-relevant EEG structure without requiring very large models or extensive pretraining.

Among recent dataset-specific studies, a multi-branch graph convolutional model with multi-head attention (MGCNA) for CHB–MIT fuses spatial distance, functional connectivity, and learnable adjacencies via attention, reporting strong patient-independent LOSO results under a 22-channel, 0.5–70 Hz, 3 s window (50% overlap) regimen [24]. For TUSZ, VQ–MTM introduces vector-quantization masked time-series pretraining with random projection and a phase-alignment module; under a clip-based protocol, it achieves competitive detection and classification performance [44]. These studies provide contemporary reference points for the datasets considered in this work.

Taken together, the literature indicates that patient-independent seizure detection remains challenging because seizure morphology, informative EEG channels, and recording conditions vary substantially across subjects. These gaps motivate the proposed subject-disjoint seizure detection framework described in Sect. 3.

3 Methodology

3.1 Datasets

We conduct all experiments on two public scalp EEG datasets: CHB–MIT and TUH Seizure Corpus (TUSZ v2.0.3). Sampling rates are handled separately for each dataset. CHB–MIT is used at its native 256 Hz, while TUSZ recordings (whose original rates vary from 250 to 1000 Hz) are resampled to 250 Hz. For each dataset, we retain the set of bipolar EEG channels common across subjects—18 channels for CHB–MIT and 20 channels for TUSZ. Dataset summaries and split statistics are shown in Tables 1 and 2.

These two datasets were selected because they are widely used public benchmarks for EEG-based seizure detection and have been adopted in several prior studies, including the representative works used for comparison in this study. Their use therefore enables evaluation under established benchmark settings while supporting comparison with existing literature. CHB–MIT provides well-annotated pediatric EEG recordings in a relatively controlled setting, whereas TUSZ provides a large-scale clinical EEG corpus with heterogeneous recordings, diverse subjects, and real-world acquisition variability. Together, these datasets allow the proposed method to be

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Table 1 CHB–MIT subject demographics and seizure counts

Patient (case)	Sex	Age (y)	Seizures
chb01	F	11	7
chb02	M	11	3
chb03	F	14	7
chb04	M	22	4
chb05	F	7	5
chb06	F	1.5	10
chb07	F	14.5	3
chb08	M	3.5	5
chb09	F	10	4
chb10	M	3	7
chb11	F	12	3
chb12	F	2	40
chb13	F	3	12
chb14	F	9	8
chb15	M	16	20
chb16	F	7	10
chb17	F	12	3
chb18	F	18	6
chb19	F	19	3
chb20	F	6	8
chb21	F	13	4
chb22	F	9	3
chb23	F	6	7
chb24	—	—	16

M = male, F = female. Age (y) denotes age in years. "—" = not reported

Table 2 TUSZ v2.0.3 (official subject-disjoint split)

Split	Subjects	EEG recordings	Duration (h)	Events (Seiz/Non-seiz)
Train	579	4,664	910.3	2,138/3,792
Dev	53	1,832	435.5	978/1,507
Eval	43	865	127.7	469/670
Total	**675**	**7,361**	**1,473.5**	**3,585/5,969**

EEG = electroencephalogram. Durations reported in hours. "Seiz/Non-seiz" denote seizure and non-seizure events

evaluated across both controlled and clinically diverse scenarios, improving the medical representativeness of the study.

3.1.1 CHB–MIT

We use the CHB–MIT pediatric scalp EEG database at its original 256 Hz sampling rate. Each recording is mapped to the 18-channel set common across subjects for spatial consistency. Table 1 reports per-case sex, age, and seizure counts; these demographics are for context, while model training and evaluation use the EEG signals and annotations.

3.1.2 TUH Seizure (TUSZ)

We use the TUH Seizure Corpus (TUSZ v2.0.3), part of a multi-year EEG collection spanning numerous subjects [11]. Original sampling rates vary from 250 to 1000 Hz; all recordings are resampled to 250 Hz for this study. Each recording is standardized to the 20-channel temporal-central-parasagittal set common across TUSZ subjects. We follow the official subject-disjoint *train/dev/eval* partition (Table 2); models are trained on *train*, tuned on *dev*, and reported on *eval*.

3.2 Preprocessing and labeling

3.2.1 Sampling and montage

Following standard practice, we retain the clinical bipolar (10–20-based) montage of each dataset [11, 13]. Within each dataset, we use the set of bipolar derivations present for every subject so that all recordings share a common channel set: 18 bipolar channels for CHB–MIT and 20 for TUSZ. Hereafter, each bipolar derivation is referred to as an EEG channel. We denote the sampling rate by f_s when needed.

Specifically, CHB–MIT montage comprises 18 bipolar pairs: FP1–F7, F7–T7, T7–P7, P7–O1; FP1–F3, F3–C3, C3–P3, P3–O1; FZ–CZ, CZ–PZ; FP2–F4, F4–C4, C4–P4, P4–O2; FP2–F8, F8–T8, T8–P8, P8–O2. TUSZ montage comprises 20 bipolar pairs following the temporal–central–parasagittal convention: FP1–F7, F7–T3, T3–T5, T5–O1; FP2–F8, F8–T4, T4–T6, T6–O2; T3–C3, C3–CZ, CZ–C4, C4–T4; FP1–F3, F3–C3, C3–P3, P3–O1; FP2–F4, F4–C4, C4–P4, P4–O2.

3.2.2 Segmentation and labels

For each recording, we extract expert-annotated seizure and non-seizure periods and segment them into fixed-length windows of $\Delta = 1$ s with 50% overlap (hop = 0.5 s).

The use of short-duration windows is consistent with the commonly adopted quasi-stationary assumption in EEG analysis, where signals are treated as approximately quasi-stationary over small temporal intervals for practical feature extraction and time–frequency analysis [2, 45]. This assumption enables the computation of statistical descriptors such as mean, standard deviation, and entropy within each segment.

However, EEG signals are inherently non-stationary. To address this, our framework does not rely solely on these statistical descriptors; instead, it combines them with EMD and DWT representations, which explicitly capture non-stationary time–frequency dynamics. This combination mitigates the limitations of the quasi-stationarity assumption and enables the model to represent both stationary and non-stationary characteristics of EEG signals.

The 1 s window with 50% overlap was used as the default configuration in the main experiments. To further assess the sensitivity of the proposed model to these segmentation choices, an additional controlled analysis evaluated window lengths of 0.5, 1, 2, and 4 s and overlap ratios of 0, 25, 50, and 75%, as reported in Sect. 5.3. We denote a window by $X \in \mathbb{R}^{C \times N}$ (rows = EEG channels, columns = time samples), where C is the common channel count in the dataset.

We use the term *episode* to denote a contiguous labeled interval, either an expert-annotated seizure interval or a contiguous non-seizure interval derived from background segments. Each episode is partitioned into fixed-length 1 s windows. Thus, an episode can contain many windows. Every window inherits the label of its parent episode.

3.3 Proposed architecture

The overall framework of the proposed seizure detection model is illustrated in Fig. 1. Starting from an input EEG segment, the system first applies a *channel attention* mechanism that adaptively reweights EEG channels to emphasize informative spatial patterns. The weighted signal is then processed through two complementary time–frequency branches, one based on empirical mode decomposition (EMD) and the other on discrete wavelet transform (DWT). Each branch produces statistical feature maps that are modeled by attention-augmented CNNs. The learned embeddings are fused and passed through a fully connected classifier at the window level. Finally, per-window probabilities are aggregated across each episode using probability averaging to obtain the final seizure/non-seizure decision. Each component is described in detail below.

3.3.1 Channel attention mechanism

To address spatial variability across subjects, we use a channel-wise attention module (Fig. 2). Given an input EEG segment $X \in \mathbb{R}^{C\times N}$ (rows = EEG channels, columns = time samples), global average pooling over time produces a channel summary:

$$\mathrm{s} = \mathrm{GAP}_t(X), \quad \mathrm{s} \in \mathbb{R}^C. \tag{1}$$

This summary passes through a bottleneck MLP with reduction ratio r:

$$\mathrm{g} = \sigma\big(W_2\,\delta(W_1\mathrm{s})\big), \tag{2}$$

where $W_1 \in \mathbb{R}^{\frac{C}{r}\times C}$, $W_2 \in \mathbb{R}^{C\times\frac{C}{r}}$, $\delta(\cdot)$ denotes ReLU, and $\sigma(\cdot)$ denotes the sigmoid function. The gating vector $\mathrm{g} \in \mathbb{R}^C$ provides adaptive per-channel weights, yielding the reweighted signal:

$$\tilde{X}_{c,t} = \mathrm{g}_c\, X_{c,t}, \qquad c = 1, 2, \ldots, C;\ t = 1, 2, \ldots, N. \tag{3}$$

This mechanism highlights channels carrying discriminative activity while down-weighting noisy or less relevant channels.

The front-end Channel Attention mechanism is used because the principal objective at this stage is to adaptively model the relative importance of the EEG channels before time–frequency feature extraction. Although its bottleneck gating operation is conceptually related to squeeze–excitation (SE) attention [7, 36], conventional SE modules are generally used to recalibrate learned feature channels within a neural network. In contrast, the proposed mechanism operates directly on the EEG channel dimension before the EMD and DWT transformations. Because the gate is applied once at the input, the same channel-weighted EEG representation is supplied to both decomposition branches, whereas attention placed after feature extraction is learned and applied independently within each branch.

CBAM combines channel and spatial attention on intermediate feature maps [8], which is useful when both learned feature-channel and spatial relationships need to be modeled jointly. Self-attention models pairwise dependencies between tokens or signal elements [9] rather than producing a single per-channel gate, and is therefore more naturally suited to recalibrating feature representations than to a front-end input-reweighting role. Channel Attention is accordingly adopted for its architectural placement: a single shared gate on the input to both decompositions. Feature-wise and spatial attention are subsequently employed within the CNN branches to model relationships among the learned time–frequency features.

3.3.2 Dual-branch time–frequency feature extraction and fusion

From the reweighted EEG $\tilde{X}$, two complementary feature sets are extracted in parallel (Fig. 3).

In the EMD branch, each channel is decomposed by empirical mode decomposition into up to seven intrinsic mode functions (IMFs), $\{m_{c,k}(t)\}_{k=1}^{7}$; if fewer IMFs occur, they are zero-padded, and if more than seven IMFs are generated, the remaining low-frequency IMF/residual components are merged into the seventh slot.

EMD produces a variable number of IMFs depending on the characteristics of the input signal [18]. Since the proposed CNN-based classifier requires a fixed-size input representation, we use a fixed upper bound of seven IMFs per channel to ensure consistent feature dimensionality across EEG windows. This choice also helps retain informative oscillatory components while avoiding an unnecessarily large feature representation.

The seven-IMF setting was selected as a practical, literature-informed, and empirically supported design choice rather than as a theoretically universal optimum. In practice, not all IMFs are equally informative; some components may capture noise, artifacts, mode-mixing effects, or slow trend-related variations [46]. Therefore, IMF selection in EEG analysis is commonly guided by signal characteristics and application-specific considerations

[2]. Several prior EEG–EMD studies have used seven IMFs or seven IMFs plus a residual component for EEG signal analysis [47–50]. During model development, we also compared nearby settings using 5, 7, and 8 IMFs, and the seven-IMF setting provided the most favorable empirical performance in our experiments.

In the DWT branch, a six-level discrete wavelet transform is applied to each channel to obtain six multi-resolution decomposition components, denoted as $\{w_{c,k}(t)\}_{k=1}^{6}$. The six-level DWT setting was selected considering the 1 s window length and the 250–256 Hz sampling rates used in this study. Because each DWT level progressively separates the signal into coarser temporal and frequency-scale components, six levels provide sufficient multi-resolution coverage while keeping the feature representation compact. This multi-resolution representation is suitable for seizure detection because ictal EEG patterns may include both short-duration high-frequency discharges and slower rhythmic changes.

From a signal-processing perspective, EMD and DWT provide complementary representations because they rely on fundamentally different decomposition principles. EMD is fully data adaptive and decomposes a non-stationary signal into intrinsic mode functions according to its local extrema and oscillatory behavior [18]. It can therefore adapt to irregular and patient-specific EEG morphologies without requiring a predefined basis function. This property is particularly useful for seizure EEG, in which transient rhythmic activity, amplitude changes, and dominant oscillatory components can vary considerably across subjects and seizure events. However, because EMD is data dependent and iterative, its decomposition can be affected by phenomena such as mode mixing and variability in the resulting IMFs [46].

In contrast, DWT analyzes the signal using a fixed wavelet basis and a structured dyadic filter bank, providing a stable and reproducible multi-resolution decomposition into progressively coarser temporal and frequency scales [25]. This makes DWT effective for representing seizure-related changes that occur simultaneously over short high-frequency intervals and longer low-frequency rhythms. However, because its basis functions and frequency partitions are predetermined, DWT is less adaptive to irregular signal-specific oscillatory structures than EMD.

Combining the two representations therefore exploits their complementary strengths: EMD provides adaptive decomposition of subject- and event-specific oscillatory activity, whereas DWT provides a consistent multi-resolution description across predefined temporal–frequency scales. The parallel branches preserve these two views separately before late fusion, allowing the classifier to integrate adaptive local oscillatory information with stable multiscale frequency information. This complementary representation is particularly relevant for patient-independent seizure detection, where ictal patterns can vary substantially in morphology, frequency content, and temporal evolution across unseen subjects.

For each component $u_{c,k}(t)$ (IMF or sub-band), six statistical descriptors are computed—mean, standard deviation, skewness, kurtosis, energy, and Shannon entropy. This results in $6 \times 7 = 42$ features per EEG channel for the EMD branch and $6 \times 6 = 36$ features per EEG channel for the DWT branch. Stacking across channels forms two compact 2-D feature maps:

$$\mathrm{F}^{(\mathrm{EMD})} \in \mathbb{R}^{C\times 42}, \qquad \mathrm{F}^{(\mathrm{DWT})} \in \mathbb{R}^{C\times 36}, \tag{4}$$

which are treated as channel–feature "images" for subsequent CNN modeling.

Each branch is modeled by a shallow CNN with two Conv–BN–ReLU blocks, followed by (i) feature-wise attention and (ii) spatial attention to emphasize salient regions. In each branch, the CNN backbone consists of two sequential convolutional blocks. Each block contains a two-dimensional convolutional layer, batch normalization, and ReLU activation. The first block uses 32 filters with a 3×3 kernel, and the second block uses 64 filters with a 3×3 kernel. The resulting feature maps are passed through feature-wise and spatial attention modules, followed by global weighted average pooling. The pooled branch embeddings from the EMD and DWT branches are concatenated and passed to a fully connected classification layer with dropout rate 0.5 and sigmoid activation.

Let $\mathrm{U} \in \mathbb{R}^{D\times H\times W}$ denote the branch feature maps. We aggregate them via global weighted average pooling (GWAP) with a learned score map $S \in \mathbb{R}^{H\times W}$:

$$W_{i,j} = \frac{\exp(S_{i,j})}{\sum_{p=1}^{H}\sum_{q=1}^{W}\exp(S_{p,q})}, \tag{5a}$$

$$z_d = \sum_{i=1}^{H}\sum_{j=1}^{W} W_{i,j}\, U_{d,i,j}, \qquad d = 1, 2, \ldots, D. \tag{5b}$$

Here, $H = C$ is the number of EEG channels and W denotes the number of features per channel ($W = 42$ for EMD and $W = 36$ for DWT). The normalized pooling weights $W_{i,j}$ aggregate activations in the GWAP operation to produce a compact descriptor $\mathrm{z} \in \mathbb{R}^D$. When the weights are uniform, GWAP reduces to standard global average pooling.

The EMD and DWT embeddings are concatenated,

$$\mathrm{e}_{\text{fused}} = [\mathrm{e}_{\text{EMD}}; \mathrm{e}_{\text{DWT}}], \tag{6}$$

then passed to a fully connected layer with dropout and sigmoid output for window-level seizure classification. This fusion preserves complementary information from the adaptive (EMD) and multiscale (DWT) views.

3.3.3 Episode-level aggregation

During inference, window-level seizure probabilities are aggregated within each contiguous episode using *soft voting* (probability averaging) (Fig. 4). Let $\{p_\ell\}_{\ell=1}^{n_j}$ denote the predicted seizure probabilities for the n_j windows that belong to episode j. We compute the episode score

$$s_j = \frac{1}{n_j}\sum_{\ell=1}^{n_j} p_\ell, \tag{7}$$

and assign the episode label by thresholding s_j. Soft voting stabilizes decisions by smoothing isolated window errors and yields reliable episode-level seizure/non-seizure predictions.

4 Training and inference protocols

We train and evaluate the proposed model under strict subject-disjoint splits. For CHB–MIT, we adopt a LOSO protocol in which each fold holds out one subject for testing and uses all remaining subjects for training; validation is drawn only from the training subjects. For TUSZ, we follow the official *train/dev/eval* partition provided along with the dataset (Table 2), training on *train*, selecting hyperparameters on *dev*, and reporting final results on *eval*.

The loss function is binary cross-entropy, which is optimized using the Adam optimizer with a learning rate of 1×10^{-3}. The model is trained with a batch size of 32 for up to 50 epochs, with early stopping based on validation F1-score to prevent overfitting. A dropout layer with a rate of 0.5 is applied in the fully connected classification stage to improve generalization. All convolutional layers use ReLU activation functions, while the final output layer uses a sigmoid activation. The complete training configuration is summarized in Table 3. To mitigate class imbalance, class-weighted binary cross-entropy loss and balanced mini-batch sampling were employed during training.

Mini-batches are sampled from the segmented windows described in Sect. 3.2 (duration Δ=1 s, 50% overlap). We sample windows so that each batch is balanced for seizure vs. non-seizure, and no subject appears in more than one split.

Table 3 Summary of training configuration and hyperparameters

Component	Configuration
Optimizer	Adam
Learning rate	1×10^{-3}
Batch size	32
Epochs	50 (with early stopping)
Loss function	Binary cross-entropy
Activation (CNN)	ReLU
Output activation	Sigmoid
Dropout rate	0.5

At test time, the same windowing is applied to obtain per-window probabilities. Within each annotated episode, these probabilities are averaged to produce a score s_j; thresholding s_j yields the episode label. This simple aggregation reduces the volatility of isolated misclassifications without modifying the classifier.

For qualitative insight, we visualize TUSZ windows using t-SNE [51]. We sample 1,000 seizure and 1,000 non-seizure 1 s windows from the *eval* split. The *raw EEG* view vectorizes each window $X \in \mathbb{R}^{C \times N}$ to $\mathbb{R}^{CN}$; the *learned embeddings* view uses the penultimate layer of our trained model (Sect. 3.3). These plots are purely qualitative and are not used for model selection or hyperparameter tuning.

We evaluate at the *episode level* using soft voting over windows. Let $y_j \in \{0, 1\}$ be the ground-truth label for episode j (1 = seizure) and $\hat{y}_j \in \{0, 1\}$ the predicted label. The counts are defined with seizure as the positive class:

- TP (true positives): seizures correctly predicted as seizure.
- FP (false positives): non-seizures incorrectly predicted as seizure.
- TN (true negatives): non-seizures correctly predicted as non-seizure.
- FN (false negatives): seizures incorrectly predicted as non-seizure.

Accuracy, Recall, and F1-score are computed as follows [34, 36]:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}, \tag{8a}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{8b}$$

$$\text{F1} = \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN}}. \tag{8c}$$

We also report the area under the receiver operating characteristic curve (AUROC) as a threshold-independent measure of discrimination [35, 38]. To improve the reliability and transparency of the reported results, performance metrics are presented as mean ± standard deviation unless otherwise stated.

To statistically assess the individual contributions of the principal model components, paired Wilcoxon signed-rank tests were performed using the subject-level F1-scores obtained from the same test subjects under the corresponding ablation configurations. The complete model (A8) was compared with A5 to assess the contribution of the front-end Channel Attention, with A6 to assess the contribution of the DWT branch, and with A7 to assess the contribution of the EMD branch. To account for the three planned component-wise comparisons, the resulting p-values were adjusted using the Holm procedure. An adjusted $p < 0.05$ was considered statistically significant.

 https://doi.org/10.1007/s00521-026-12518-w

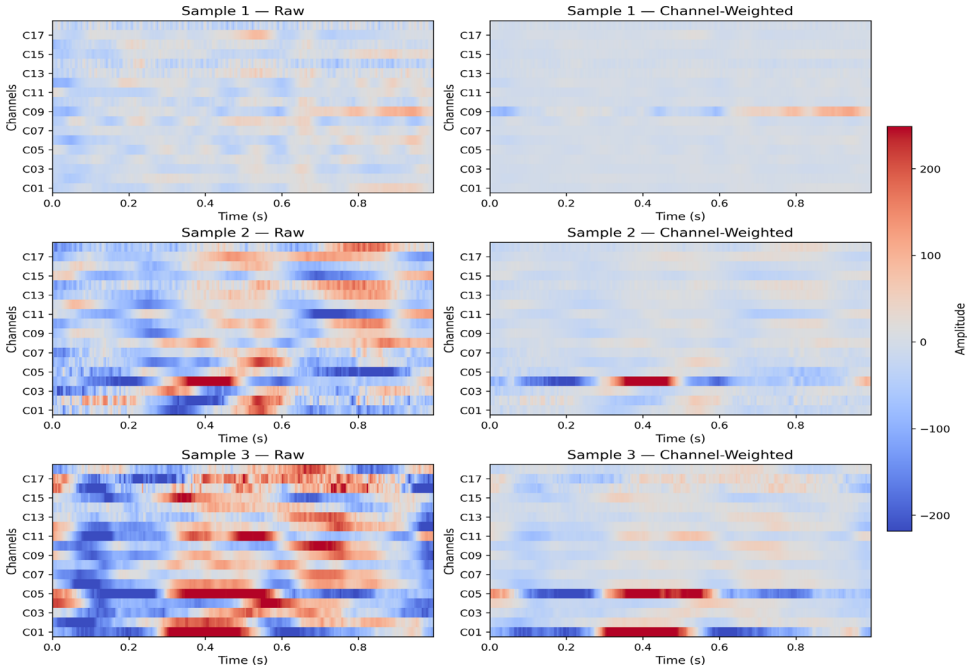


Fig. 5 Representative samples illustrating the effect of channel-wise weighting. Each row shows a different 1 s EEG window (low/medium/high activity). Left column: raw EEG; right column: channel-weighted EEG using the learned per-window gains (Sect. 3.3). Channel weighting suppresses weak/background channels and emphasizes dominant ones, increasing spatial contrast through input-dependent learned gains.

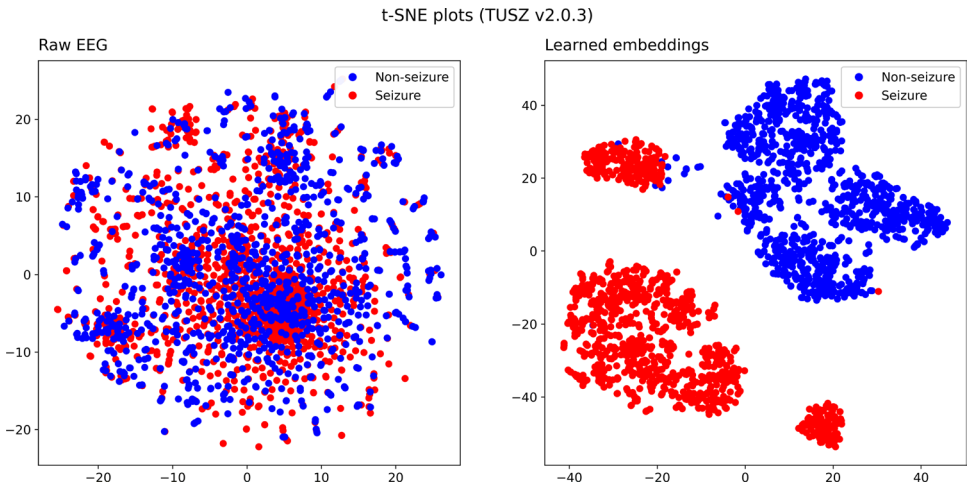


Fig. 6 t-SNE on TUSZ v2.0.3 (eval split). Left: raw EEG windows. Right: learned embeddings from the pen-ultimate layer of our model (Sect. 3.3). Both plots show 1,000 seizure (red) and 1,000 non-seizure (blue) samples.

5 Results and ablations

We begin with qualitative evidence that channel-wise attention enhances spatial contrast and yields separable embeddings (Figs. 5 and 6). We then report ablation studies on CHB–MIT and TUSZ (Tables 4 and 5) that isolate the contributions of each component: channel attention, dual-branch EMD–DWT fusion, and intra-CNN attention. These ablation results are also visualized using standard-deviation error bars in Fig. 7 to provide a clearer view of variability and reliability in the reported performance. We then examine the sensitivity of the model to window length and overlap ratio (Sect. 5.3), report a continuous-recording evaluation using clinically relevant metrics (Sect. 5.4), and analyze the false-positive, false-negative, and failure cases (Sect. 5.5). We further evaluate robustness to additive Gaussian noise at different SNR levels (Sect. 5.6, Tables 9 and 10) and analyze the computational complexity and measured runtime of the pipeline (Section 5.7). Finally, we compare the full model with representative literature studies on CHB–MIT and TUSZ (Sect. 5.8, Tables 13 and 14).

Table 4 Ablation study on CHB–MIT under the LOSO protocol. The configurations isolate the contributions of the EMD and DWT branches, front-end Channel Attention, and intra-CNN attention. Metrics are computed at the episode level and reported as mean ± standard deviation across LOSO folds

ID	Configuration	Accuracy (%)	AUROC	F1 (%)	Recall (%)
A1	EMD branch only	84.28 ± 8.23	0.87 ± 0.08	85.12 ± 9.28	85.67 ± 8.96
A2	DWT branch only	80.56 ± 9.45	0.83 ± 0.07	80.68 ± 8.87	81.35 ± 9.29
A3	EMD + DWT without any attention	89.60 ± 9.61	0.89 ± 0.09	90.10 ± 8.62	90.21 ± 9.44
A4	EMD + DWT with Channel Attention only	91.22 ± 8.11	0.92 ± 0.06	90.21 ± 8.97	90.60 ± 8.73
A5	EMD + DWT with intra-CNN attention only	90.39 ± 8.38	0.91 ± 0.07	90.14 ± 9.15	90.92 ± 8.78
A6	Complete model without DWT	84.95 ± 8.60	0.88 ± 0.07	85.24 ± 9.29	85.49 ± 8.86
A7	Complete model without EMD	81.33 ± 8.80	0.84 ± 0.07	81.64 ± 9.01	81.93 ± 8.73
A8	Complete proposed model	**91.63 ± 9.29**	**0.93 ± 0.07**	**90.25 ± 10.51**	**91.34 ± 10.13**

Table 5 Ablation study on TUSZ v2.0.3 using the official subject-disjoint eval split. The configurations isolate the contributions of the EMD and DWT branches, front-end Channel Attention, and intra-CNN attention. Metrics are computed at the episode level and reported as mean ± standard deviation across evaluation subjects

ID	Configuration	Accuracy (%)	AUROC	F1 (%)	Recall (%)
A1	EMD branch only	85.67 ± 5.31	0.86 ± 0.06	84.32 ± 4.18	86.20 ± 4.78
A2	DWT branch only	83.24 ± 6.28	0.83 ± 0.07	83.55 ± 4.67	84.72 ± 3.71
A3	EMD + DWT without any attention	88.35 ± 4.39	0.90 ± 0.05	89.00 ± 4.26	88.84 ± 3.98
A4	EMD + DWT with Channel Attention only	90.13 ± 4.72	0.92 ± 0.05	90.43 ± 4.08	90.61 ± 4.47
A5	EMD + DWT with intra-CNN attention only	89.21 ± 4.63	0.91 ± 0.05	89.39 ± 4.06	89.74 ± 4.43
A6	Complete model without DWT	86.12 ± 5.25	0.87 ± 0.06	86.57 ± 4.34	86.73 ± 4.87
A7	Complete model without EMD	83.86 ± 5.42	0.84 ± 0.06	84.11 ± 4.63	84.48 ± 5.09
A8	Complete proposed model	**92.87 ± 4.85**	**0.93 ± 0.05**	**92.65 ± 3.94**	**92.58 ± 4.66**

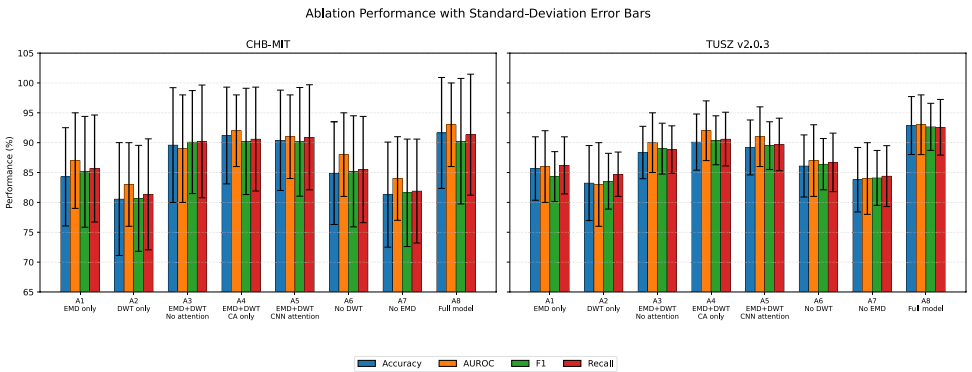


Fig. 7 Ablation performance with standard-deviation error bars on CHB–MIT and TUSZ. Bars show the mean episode-level performance, and error bars represent one standard deviation across LOSO folds (one held-out subject per fold) for CHB–MIT and across evaluation subjects for TUSZ. AUROC values are multiplied by 100 only for visualization on the same scale as accuracy, F1-score, and recall; AUROC is otherwise reported on a [0, 1] scale. In the configuration labels, CA denotes front-end Channel Attention

5.1 Qualitative assessment

We first illustrate the qualitative effect of channel-wise attention in Fig. 5. We show three representative 1 s windows (rows) spanning low/medium/high activity. For each window, we display the raw EEG heatmap and the corresponding channel-weighted version (columns). Across all three examples, the attention layer attenuates less-informative channels and enhances a small subset carrying stronger structure, yielding better spatial contrast. The effect is most pronounced for higher-activity windows, but remains visible at moderate activity as well.

Next, Fig. 6 shows t-SNE visualizations of raw EEG windows (left) and learned model embeddings (right). The raw windows form a largely intermixed cloud, reflecting the effect of clinical heterogeneity and artifacts. In contrast, the learned embedding exhibits clear seizure/non-seizure grouping, indicating that the representation suppresses nuisance variability while preserving ictal structure. This visualization is illustrative; quantitative evaluations are reported in the subsequent subsections.

5.2 Ablation studies

We evaluate eight model configurations (Tables 4 and 5) to examine the individual and combined contributions of the EMD branch, DWT branch, front-end Channel Attention, and intra-CNN attention. A1 and A2 evaluate the EMD-only and DWT-only branches, respectively, with both the front-end Channel Attention and intra-CNN attention disabled. A3 combines the EMD and DWT branches without either attention mechanism. A4 combines EMD and DWT with only the front-end Channel Attention, whereas A5 combines EMD and DWT with only the intra-CNN attention mechanism. A6 corresponds to the complete attention-equipped architecture with the DWT branch removed, while A7 corresponds to the complete attention-equipped architecture with the EMD branch removed. A8 represents the complete proposed model containing both EMD and DWT branches together with front-end Channel Attention and intra-CNN attention.

Combining the two time–frequency representations without attention (A3) improves accuracy over both single-branch configurations, EMD only (A1) and DWT only (A2), on CHB–MIT and TUSZ (Tables 4 and 5). These results suggest that the adaptive EMD representation and the structured multi-resolution DWT representation provide complementary information.

Introducing the attention mechanisms individually further improves the dual-branch configuration. On both datasets, adding the front-end Channel Attention (A4) to A3 yields the larger gain in mean accuracy: 1.62 percentage points on CHB–MIT and 1.78 percentage points on TUSZ, compared with corresponding gains of 0.79 and 0.86 percentage points for intra-CNN attention alone (A5). Both mechanisms therefore provide additional benefit when applied to the dual EMD–DWT representation.

The branch-removal experiments further demonstrate the importance of both time–frequency representations within the complete attention-equipped architecture. Relative to the complete model (A8), removing EMD (A7) reduces mean accuracy by 10.30 percentage points on CHB–MIT and 9.01 percentage points on TUSZ, whereas removing DWT (A6) reduces it by 6.68 and 6.75 percentage points, respectively. The consistently larger reduction after removing EMD suggests that the EMD branch makes a stronger contribution to mean accuracy within the complete architecture.

Overall, A8 achieves the highest accuracy, AUROC, F1-score, and recall on both datasets. These results support the complementary roles of EMD, DWT, and the two attention mechanisms.

Statistical significance analysis of the F1-score was further performed for the component-wise ablation comparisons. For CHB–MIT, the Holm-adjusted p-values for Channel Attention (A8 vs. A5), DWT (A8 vs. A6), and EMD (A8 vs. A7) were $p = 0.021$, $p = 0.002$, and $p = 0.001$, respectively. For TUSZ, the corresponding adjusted p-values were $p = 0.018$, $p = 0.004$, and $p = 0.003$, respectively.

5.3 Sensitivity to window length and overlap ratio

The sensitivity analysis (Tables 6 and 7) shows that the 1 s window with 50% overlap provides the most favorable overall performance on both datasets. Each configuration was retrained and evaluated under the same subject-disjoint protocol, with all other experimental settings kept unchanged.

For the window-length experiment, reducing the window to 0.5 s decreases performance, suggesting that very short segments may not contain sufficient temporal context to represent seizure-related dynamics reliably. Increasing the window length to 2 s or 4 s also results in lower overall performance than the 1 s setting, indicating

Table 6 Effect of window length on the episode-level performance of the proposed model with the overlap ratio fixed at 50%. All other experimental settings were kept unchanged. Results are reported as mean ± standard deviation

Window length (s)	Accuracy (%)	AUROC	F1 (%)	Recall (%)
CHB–MIT (LOSO)				
0.5	85.32 ± 7.32	0.87 ± 0.06	84.71 ± 11.47	85.51 ± 10.82
1.0	91.63 ± 9.29	0.93 ± 0.07	90.25 ± 10.51	91.34 ± 10.13
2.0	87.32 ± 9.73	0.89 ± 0.09	86.27 ± 9.89	87.35 ± 10.64
4.0	86.87 ± 9.59	0.90 ± 0.08	86.05 ± 10.78	87.00 ± 9.89
TUSZ v2.0.3 (Eval)				
0.5	88.23 ± 7.10	0.88 ± 0.06	87.98 ± 8.05	87.95 ± 7.80
1.0	92.87 ± 4.85	0.93 ± 0.05	92.65 ± 3.94	92.58 ± 4.66
2.0	91.43 ± 5.31	0.92 ± 0.05	91.19 ± 4.52	91.15 ± 5.04
4.0	90.12 ± 6.27	0.91 ± 0.06	89.88 ± 5.79	89.83 ± 6.12

The overlap ratio was fixed at 50% for all window-length configurations

Table 7 Effect of overlap ratio on the episode-level performance of the proposed model with the window length fixed at 1 s. All other experimental settings were kept unchanged. Results are reported as mean ± standard deviation

Overlap (%)	Accuracy (%)	AUROC	F1 (%)	Recall (%)
CHB–MIT (LOSO)				
0	84.11 ± 9.19	0.83 ± 0.08	84.48 ± 10.13	85.22 ± 9.81
25	87.18 ± 7.46	0.87 ± 0.06	86.92 ± 8.20	87.11 ± 8.03
50	91.63 ± 9.29	0.93 ± 0.07	90.25 ± 10.51	91.34 ± 10.13
75	89.86 ± 9.51	0.91 ± 0.08	88.71 ± 10.26	89.81 ± 9.98
TUSZ v2.0.3 (Eval)				
0	86.24 ± 6.52	0.88 ± 0.06	85.91 ± 6.81	86.52 ± 6.23
25	89.49 ± 6.12	0.91 ± 0.06	89.12 ± 6.44	89.33 ± 6.11
50	92.87 ± 4.85	0.93 ± 0.05	92.65 ± 3.94	92.58 ± 4.66
75	91.91 ± 5.01	0.92 ± 0.05	90.54 ± 5.86	91.76 ± 5.43

The window length was fixed at 1 s for all overlap-ratio configurations

that longer segments may combine heterogeneous temporal patterns and reduce the temporal specificity of the extracted features.

A similar trend is observed for the overlap ratio (Table 7). Performance at 50% overlap is higher than at either 0% or 25% overlap, indicating that moderate overlap provides denser temporal sampling and reduces the likelihood of losing discriminative activity near window boundaries. Increasing the overlap further to 75% does not provide an additional improvement and instead produces a slight reduction in performance. These results are consistent with the 1 s window and 50% overlap adopted for the main experiments, which offers a favorable balance between temporal resolution and contextual information across both CHB–MIT and TUSZ.

5.4 Continuous-recording clinical evaluation

To assess the proposed model using clinically relevant continuous-monitoring metrics, we additionally evaluated the trained model by processing the complete test EEG recordings sequentially. The recordings were processed using 1 s EEG windows with 50% overlap (0.5 s hop), consistent with the configuration used in the main experiments. The windows were supplied to the trained model in their original temporal order without using seizure boundaries during prediction.

A window was labeled seizure-positive when its predicted seizure probability exceeded a fixed decision threshold of 0.5. Consecutive seizure-positive windows were merged into a single alarm, so that one continuous detection is counted as a single event rather than as multiple overlapping windows.

Expert seizure annotations were used only after prediction to classify the resulting alarms. An alarm was counted as a correct detection if it overlapped the expert-annotated seizure interval by at least one window, and each annotated seizure was counted as detected at most once. Alarms falling entirely within non-seizure EEG were counted as false alarms. On this basis, we computed the false alarm rate per hour (FAR/h) and seizure detection latency, defined as the interval between the annotated seizure onset and the first corresponding correct alarm. The resulting values for CHB–MIT and TUSZ are summarized in Table 8.

This continuous-recording evaluation complements the episode-level results by characterizing model behavior during sequential processing of complete test recordings without prior knowledge of seizure boundaries.

5.5 False-positive, false-negative, and failure-case analysis

To further examine the failure modes of the proposed model, we analyzed the false-positive (FP) and false-negative (FN) predictions obtained on the held-out test data. On CHB–MIT, the model produced 24 false-positive episodes and 17 false-negative seizure episodes across the LOSO test folds. On TUSZ, 34 false-positive episodes and 32 false-negative seizure episodes were observed on the official evaluation split. These counts represent pooled episode-level classification errors across the held-out test subjects.

Qualitative inspection of the error cases against the corresponding raw EEG and expert annotations indicated that false positives were commonly associated with muscle artifacts, electrode-pop artifacts, and abrupt high-amplitude transient activity, which can produce time–frequency characteristics resembling ictal activity. False negatives, in contrast, were more frequently associated with seizures shorter than 10 s, low-amplitude ictal discharges, or spatially focal seizures. In these cases, the discriminative seizure activity was weak or confined to a limited subset of EEG channels.

Overall, the failure-case analysis indicates that the major challenges are seizure-like non-ictal transients that increase false-positive detections and weak, short, or spatially restricted ictal activity that can lead to missed detections. These findings suggest that future improvements may benefit from artifact-aware temporal modeling and sequence-level decision mechanisms that incorporate information across consecutive EEG windows.

5.6 Robustness to additive noise

To assess sensitivity to signal distortions, we introduced additive Gaussian noise to the test data at SNR levels of 20 dB, 10 dB, 5 dB, and 0 dB without retraining the model. Additive Gaussian noise was used as a standard perturbation model commonly adopted for evaluating robustness in EEG signal analysis. The 20 dB, 10 dB, and 5 dB conditions represent increasing levels of signal corruption, while 0 dB was included as an extreme stress-test condition in which the added noise power is equal to the EEG signal power. Since the original EEG recordings were collected in real clinical environments, they already contain various physiological and environmental artifacts. Therefore, the added Gaussian noise serves as an additional controlled perturbation for evaluating the robustness of the proposed framework under increasing signal corruption. As shown in Tables 9 and 10, overall performance generally decreases as the SNR decreases, indicating stable behavior and graceful degradation under increasing additive-noise corruption. These results provide an initial indication of the robustness of the proposed framework under varying EEG signal quality.

Table 8 Continuous-recording evaluation of the proposed model using clinically relevant seizure-detection metrics

Dataset	FAR/h	Detection latency (s)
CHB–MIT (LOSO)	0.30 ± 0.07	3.22 ± 1.08
TUSZ v2.0.3 (Eval)	0.39 ± 0.12	4.51 ± 1.32

Note: FAR/h is the number of false-alarm events per hour of non-seizure EEG. Consecutive seizure-positive windows are merged into one alarm. FAR/h is reported as mean ± standard deviation across test subjects. Detection latency is measured from the annotated seizure onset to the first corresponding detection and is reported as mean ± standard deviation across correctly detected seizure events

Table 9 Robustness analysis under additive Gaussian noise on CHB–MIT LOSO test folds (no retraining)

SNR (dB)	Accuracy (%)	AUROC	F1 (%)	Recall (%)
Original EEG (no noise)	91.63 ± 9.29	0.93 ± 0.07	90.25 ± 10.51	91.34 ± 10.13
20 dB	91.27 ± 8.84	0.92 ± 0.04	88.94 ± 9.25	91.02 ± 9.71
10 dB	89.34 ± 9.61	0.90 ± 0.09	87.32 ± 9.56	89.29 ± 8.95
5 dB	86.62 ± 9.98	0.89 ± 0.08	84.82 ± 10.12	87.46 ± 9.24
0 dB	82.72 ± 8.79	0.83 ± 0.05	81.39 ± 9.86	82.78 ± 9.63

Table 10 Robustness analysis under additive Gaussian noise on the TUSZ eval set (no retraining)

SNR (dB)	Accuracy (%)	AUROC	F1 (%)	Recall (%)
Original EEG (no noise)	92.87 ± 4.85	0.93 ± 0.05	92.65 ± 3.94	92.58 ± 4.66
20 dB	90.98 ± 4.89	0.93 ± 0.04	91.69 ± 4.13	91.68 ± 4.39
10 dB	89.55 ± 4.66	0.91 ± 0.07	90.38 ± 3.87	91.03 ± 4.58
5 dB	87.91 ± 3.95	0.90 ± 0.04	89.52 ± 3.68	88.72 ± 3.76
0 dB	83.84 ± 4.57	0.86 ± 0.05	84.87 ± 4.19	83.54 ± 3.97

Table 11 Approximate theoretical computational complexity of the major stages of the proposed framework for one EEG window

Processing stage	Approximate complexity
Input Channel Attention and signal reweighting	$\mathcal{O}(CN + C^2/r)$
EMD decomposition	$\mathcal{O}(CKSN)$
Statistical features from the IMFs	$\mathcal{O}(CKN)$
DWT decomposition and statistical features	$\mathcal{O}(CQN) \approx \mathcal{O}(CN)$
Dual CNN branches	$\mathcal{O}\left(\sum_{b=1}^{2}\sum_{l=1}^{L_b} H_{b,l} W_{b,l} \kappa_{b,l}^2 F_{b,l-1} F_{b,l}\right)$
Fusion and binary classification	$\mathcal{O}(d_{\text{fused}})$

5.7 Computational complexity and runtime analysis

We further analyze the theoretical computational cost of the main stages of the proposed framework. Let C denote the number of EEG channels, N the number of samples in each EEG window, K the number of retained intrinsic mode functions (IMFs), S the average number of sifting iterations required to extract each IMF, Q the length of the DWT analysis filters, and r the reduction ratio of the input Channel Attention module. In the present implementation, $C = 18$ and $N = 256$ for CHB–MIT, whereas $C = 20$ and $N = 250$ for TUSZ. The maximum number of retained IMFs is $K = 7$, and the DWT decomposition depth is fixed at six levels.

The input Channel Attention module performs temporal averaging, bottleneck transformation, and signal reweighting. Its computational cost is therefore $\mathcal{O}(CN + C^2/r)$. Assuming that extrema detection and envelope construction have approximately linear cost with respect to N during each sifting iteration, the approximate cost of EMD decomposition is $\mathcal{O}(CKSN)$. The value of this term is signal dependent because the number of IMFs and the number of sifting iterations can vary across EEG windows.

For a decimated DWT, the number of coefficients decreases by approximately one half at each successive level. Consequently, the total cost across the fixed six decomposition levels is $\mathcal{O}(CQN)$, which reduces to $\mathcal{O}(CN)$ when the wavelet filter length is treated as a constant. Computation of the statistical descriptors requires $\mathcal{O}(CKN)$ for the EMD branch and approximately $\mathcal{O}(CN)$ for the DWT branch. The dominant cost of the two CNN branches is determined by their convolutional layers. Table 11 summarizes the theoretical cost of the major processing stages.

In the CNN expression, b identifies the EMD or DWT branch, L_b is the number of convolutional layers in branch b, $H_{b,l}$ and $W_{b,l}$ are the spatial dimensions of the feature map at layer l, $\kappa_{b,l}$ is the convolutional kernel size, and $F_{b,l-1}$ and $F_{b,l}$ are the numbers of input and output feature maps, respectively. The computational costs of the feature-wise attention, spatial attention, and global weighted average pooling operations are lower than the

Table 12 Average processing time of the proposed pipeline per 1 s EEG window on the reported workstation

Processing stage	Time per 1 s window (ms)
EMD-based feature extraction	115.95
DWT-based feature extraction	0.35
CNN inference	0.41
Total processing time	116.71

Table 13 Comparison with representative subject-independent CHB–MIT seizure detection studies

Study	Method	Accuracy (%)	AUROC	F1 (%)	Recall (%)
Zhang et al. [20]	Patient-adversarial learning with spatio-temporal EEG augmentation	NA	0.9098	NA	88.04
Li et al. [24]	Multi-branch GCN with multi-head attention	89.45	0.89	87.67	84.72
Ali et al. [43]	Random Forest with event-level post-processing	NA	NA	NA	75.34
Liu et al. [52]	Dual-stream CNN using S-transform phase and power spectra	92.10	0.85	NA	79.59
Proposed model	**Channel attention + EMD–DWT + attention-CNN**	**91.63 ± 9.29**	**0.93 ± 0.07**	**90.25 ± 10.51**	**91.34 ± 10.13**

NA = not available/not reported

dominant convolutional cost because they operate on the compact intermediate feature maps. Here, d_{fused} denotes the dimension of the concatenated EMD–DWT embedding supplied to the binary classifier.

Accordingly, the approximate per-window complexity of the complete framework can be expressed as

$$\mathcal{O}\left(CN + \frac{C^2}{r} + CKSN + CKN + CQN + \mathcal{C}_{\text{CNN}} + d_{\text{fused}}\right), \tag{9}$$

where $\mathcal{C}_{\text{CNN}}$ denotes the convolutional cost defined in Table 11. Because C, K, the DWT depth, and the CNN architecture are fixed in the present framework, EMD is expected to be the most computationally demanding stage because of its iterative and data-adaptive sifting process.

To complement the theoretical analysis, we measured the average processing time of the proposed pipeline per 1 s EEG window. Runtime was measured on a workstation equipped with an 11th-generation Intel Core i5 CPU, 32 GB RAM, and an NVIDIA GeForce RTX 4060 GPU, running Linux Mint 21.2 with Python 3.11.5 and TensorFlow 2.18.0. Execution time was measured separately for EMD-based feature extraction, DWT-based feature extraction, and CNN inference, and each reported value is the mean over 2,500 randomly selected 1 s EEG windows. The reported total is the sum of the three stage-wise measurements.

As shown in Table 12, EMD-based feature extraction accounts for most of the measured processing time, which is consistent with its iterative theoretical cost of $\mathcal{O}(CKSN)$. In contrast, DWT feature extraction has approximately linear complexity for a fixed wavelet and decomposition depth, while CNN inference accounts for only a small proportion of the total processing time. The complete pipeline requires 116.71 ms to process a 1 s EEG window. Under the adopted 50% overlap, successive windows advance by 0.5 s; therefore, the measured processing time remains below the 500 ms window-advancement interval on the reported hardware. These findings provide an initial indication of computational feasibility, although further implementation-level optimization and continuous-stream validation would be required before clinical deployment.

5.8 Comparison

We compare the proposed model with representative literature studies on CHB–MIT and TUSZ (Tables 13 and 14). For CHB–MIT, the comparison includes subject-independent, cross-patient, and cross-subject seizure

Table 14 Comparison with representative TUSZ seizure detection studies

Study	Method	Accuracy (%)	AUROC	F1 (%)	Recall (%)
Gui et al. [44]	VQ–MTM with random projection and phase alignment	NA	0.904	NA	NA
Huang et al. [53]	Self-supervised AFTA Transformer	NA	0.891	NA	NA
Proposed model	**Channel attention + EMD–DWT + attention-CNN**	**92.87 ± 4.85**	**0.93 ± 0.05**	**92.65 ± 3.94**	**92.58 ± 4.66**

NA = not available/not reported

detection studies. For TUSZ, the comparison includes recent TUSZ-based seizure detection studies, including self-supervised and Transformer-based approaches.

Overall, the proposed model achieves competitive performance under strict subject-disjoint evaluation, while using compact EMD–DWT representations, channel-wise attention, and episode-level probability averaging.

In the present study, our focus is to develop a compact and domain-informed patient-independent seizure detector using structured, interpretable EMD–DWT time–frequency descriptors and lightweight attention-augmented CNNs. Accordingly, while ViT, Swin Transformer, DenseNet, and recent EEG Transformer/foundation-model studies are discussed as relevant literature to place the proposed method in the broader deep learning context, the quantitative comparison is limited to seizure-detection models with similar datasets, task definitions, evaluation protocols, and reported metrics, so that the comparison remains fair and meaningful.

For EEG analysis, there are various additional aspects for adapting Transformer-based models through EEG-specific input formulations and training strategies, such as channel-wise tokenization, raw-signal segmentation, masked modeling, self-supervised pretraining, or cross-dataset learning [30–33]. For a fair comparison, such a study would ideally need to consider differences in input representation, pretraining of generic backbones, attention mechanisms, and training strategies, which is beyond the scope of this work. However, a systematic matched comparison that considers these factors would be a useful direction for adapting such generic architectures to EEG applications.

For this reason, Tables 13 and 14 focus on seizure-detection studies with closer methodological settings and reported metrics.

6 Discussion

6.1 Main findings

This work focuses on the patient-independent setting relevant to deployment. By combining (i) a channel-wise attention front end that reweights EEG channels, (ii) dual-branch EMD–DWT summaries, and (iii) attention within lightweight CNN backbones, the model emphasizes informative EEG channels and learns robust representations from compact time–frequency descriptors, while episode-level averaging (soft voting) stabilizes predictions. The design preserves the interpretability and efficiency of classical features, and attention is designed to help address inter-subject variability. The ablations indicate that the front-end attention, dual-branch fusion, and intra-CNN attention each contribute to the observed performance improvements.

However, there is scope for further improvement and extension. The current study focuses on binary seizure/non-seizure detection using fixed 1 s windows and episode-level averaging. Continuous-recording performance is additionally assessed using FAR/h and detection latency (Sect. 5.4). False positives were commonly associated with artifacts, whereas missed detections were commonly associated with short-duration, low-amplitude, or spatially focal seizures.

Cross-subject variability remains a fundamental challenge because seizure characteristics vary considerably across patients in terms of spatial distribution, amplitude, and temporal evolution. This variability can reduce generalization when the model encounters previously unseen subject-specific patterns.

Recent work in medical AI has emphasized interpretable hybrid pipelines that integrate mathematically grounded feature extractors with deep learning architectures. Such approaches, including moment-based feature representations combined with CNNs [39–42], highlight the importance of preserving domain-relevant structure while enabling data-driven learning. These ideas align with our use of EMD and DWT, which provide interpretable time–frequency representations that are subsequently modeled using attention-based neural networks.

6.2 Clinical implications

From a clinical perspective, the continuous-recording results in Sect. 5.4 are more informative than the episode-level metrics alone, because they describe how the model would behave during uninterrupted monitoring. The higher FAR/h and longer detection latency observed on TUSZ may partly reflect the greater heterogeneity of that corpus, including differences in subjects, acquisition systems, and recording conditions.

Although these retrospective continuous-recording results are encouraging, they should not be interpreted as a substitute for prospective real-time clinical validation. Practical deployment would additionally require evaluation under prospective streaming conditions, across independent clinical centers and recording systems, together with further investigation of alarm post-processing, robustness to artifacts, and computational optimization.

6.3 Limitations and future directions

Several limitations of the present study should be acknowledged. First, no independent external hospital cohort was available, so differences in patient populations, acquisition hardware, montage, and clinical practice may affect generalization beyond CHB–MIT and TUSZ. External multicenter validation, and ultimately prospective evaluation on continuously acquired clinical EEG, remains necessary.

Second, the framework performs binary seizure/non-seizure detection and does not distinguish among seizure types. Extending it to multi-class classification would provide more clinically informative outputs.

Third, the model operates on fixed-duration windows followed by probability aggregation. The continuous-recording evaluation reported in Sect. 5.4 provides FAR/h and detection-latency estimates, but the framework still does not explicitly model the temporal evolution of seizure onset and offset. Sequence-aware architectures could further improve boundary localization and the detection of short or weakly expressed events.

Fourth, seizure-like artifacts and weak or spatially restricted ictal patterns remain the main sources of false alarms and missed detections. Artifact-aware modeling, signal-quality assessment, and adaptive alarm generation could reduce these further.

Finally, EMD dominates the computational cost of the pipeline. Although the measured processing time remains below the window-advancement interval, accelerated or parallel EMD, alternative adaptive decompositions, or learned approximations could reduce it.

Taken together, these limitations define priorities for broader validation and further methodological development.

7 Conclusion

We presented a channel-attentive dual-branch seizure detector for the patient-independent setting. Channel-wise reweighting at the input, complementary EMD–DWT representations modeled by attention-augmented CNNs, and episode-level probability averaging together improve cross-subject generalization on CHB–MIT and TUSZ under strict subject-disjoint evaluation, while keeping the model compact and its features interpretable. The continuous-recording evaluation indicates that this behavior carries over to sequential processing of complete recordings, although external multicenter data and prospective streaming validation remain necessary before clinical use.

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Author contributions Hrishikesh Tiwary led the research design, implementation, data collection, analysis, manuscript preparation, and manuscript review. Arnav Bhavsar contributed to analysis, manuscript review, and supervision. Muralikrishna H assisted in implementation and manuscript review.

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Data availability The datasets analyzed during this study are publicly available. The CHB–MIT Scalp EEG Database (version 1.0.0), which contains continuous EEG recordings and expert seizure onset/offset annotations from pediatric subjects, is available from PhysioNet (https://physionet.org/content/chbmit/1.0.0/; DOI: 10.13026/C2K01R). The TUH EEG Seizure Corpus (TUSZ v2.0.3), which contains manually annotated seizure events, is available through the Neural Engineering Data Consortium website (https://isip.piconepress.com/projects/nedc/html/tuh_eeg/index.shtml#c_tusz). Access to these datasets may require agreeing to the respective repository terms. All recordings were fully anonymized by the dataset providers prior to public release, and no new data were generated in this study.

Declarations

Conflicts of interest The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Ethical approval Because this study involves secondary analysis of fully anonymized public datasets, institutional ethics approval and informed consent were not required.

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